

Customer Sales & Churn Analysis

1. Project Overview

This project analyzes sales transaction data and customer churn data using Python, Pandas, and Matplotlib. The objective is to understand sales performance, identify strong products and regions, measure customer churn, and provide actionable business recommendations.

2. Objectives

- Analyze overall sales performance and transaction volume.
- Compare product-wise and region-wise sales.
- Identify important sales trends.
- Calculate customer churn rate.
- Analyze churn by contract, payment method, billing, tenure and charges.
- Provide business insights and recommendations.

3. Sales Analysis

Metric	Value
Total Transactions	100
Total Quantity Sold	478
Total Sales	12,365,048
Average Sales per Transaction	123,650.48
Highest Sales Date	2024-01-16
Highest Daily Sales	373,932

Product-wise Sales

Product	Sales
Laptop	3,889,210
Tablet	2,884,340
Phone	2,859,394
Headphones	1,384,033
Monitor	1,348,071

Regional Sales

Region	Sales
North	3,983,635
South	3,737,852
East	2,519,639
West	2,123,922

4. Customer Churn Analysis

Metric	Value
Total Customers	500
Churned Customers	53
Non-Churned Customers	447
Churn Rate	10.6%
Avg. Tenure - Non-Churned	40.15 months
Avg. Tenure - Churned	6.00 months
Avg. Monthly Charges - Non-Churned	111.72
Avg. Monthly Charges - Churned	129.77

Churn by Contract

- Month-to-month: 20.59%
- One year: 4.30%
- Two year: 6.94%

Churn by Payment Method

- Bank Transfer: 6.92%
- Credit Card: 13.48%
- Electronic Check: 11.04%

5. Key Insights

- Laptops are the strongest-performing product in the sales dataset.
- North is the strongest region, while West records the lowest sales.
- The overall customer churn rate is 10.6%.
- Month-to-month customers have the highest churn rate among contract types.
- In this dataset, churn is concentrated among customers with 0–12 months of tenure.
- Churned customers have higher average monthly charges than non-churned customers.

6. Recommendations

- Promote high-performing products while improving lower-performing regions.
- Encourage month-to-month customers to adopt longer-term contracts through suitable retention offers.
- Strengthen onboarding and engagement during the first 12 months.
- Review pricing and plan value for customers with higher monthly charges.
- Use targeted retention campaigns and early-warning churn monitoring.

7. Conclusion

The analysis provides a combined view of sales performance and customer retention. The results can support product planning, regional sales strategy, and targeted customer retention initiatives.